

EMR Module Specification & Functional Guide

SYSTEM REFERENCE	TARGET MODULE	DOCUMENT VERSION	VALIDATION STATUS
SimplexWorld Master v9.3 / v9.5	Electronic Medical Records (EMR)	v2.1 Production Baseline	✔ 100% Passed (Gates A–E)

1. Executive Overview & Portal Hub

The **Simplex HIMS EMR System** operates as a high-reliability, HIPAA-compliant clinical operations hub. It interconnects outpatient doctor consultations, inpatient nursing ward care, electronic medication administration records (eMAR), discharge reconciliation, diagnostic ordering (CPOE), and real-time clinical governance.

The central clinical waiting list and navigation hub (/MasterV9.3/viewEMRWaitingList/1) coordinates clinical workflows across 7 core screens and 16 specialized clinical tabs.

2. Complete Screen-by-Screen Functional Guide

1. EMR Workstation (Clinical Workstation)

DOCTOR / NURSE ACTIVE CARE

Route: /MasterV9.3/emrWorkstation • State: patientemr.*

Primary Function: Complete electronic medical record and consultation workspace for physicians, surgeons, and specialists.

- **Persistent Patient Identity Banner:** Displays Patient Name, MRN, Age, Gender, Blood Group, Active Visit/Encounter Number, Ward/Bed number, and Attending Consultant with persistent color-coded allergy badges (NKDA vs. active allergen pills).
- **Chief Complaint & History of Present Illness (HPI):** Structured capture of primary complaints, onset duration, pain scale (1–10), and 14-system Review of Systems (ROS).
- **Vital Signs & Biometric Trajectory:** Captures BP (Sys/Dia), Pulse, Temp (°F), Resp Rate, SpO2 (%), Height, Weight, Blood Glucose, and Pain Score. Auto-calculates metric BMI and flags abnormal physiological values against clinical reference thresholds (VITAL_RANGES).
- **ICD-10 Diagnostic Problem List:** Direct autocomplete search of standard ICD-10 diagnostic codes, categorized into *Provisional* (working), *Final* (confirmed), and *Differential* diagnoses.
- **CPOE (Clinical Orders Entry):** Order entry for Laboratory, Radiology/Imaging, Procedures, and Consultations with STAT (Immediate), Urgent, and Routine priority flags with mandatory clinical indication capture.
- **Electronic Prescriptions (Rx) & Dosing Math:** Formulary integration calculating exact total quantities via $\text{Quantity} = \text{Dose Multiplier} \times \text{Daily Intakes} \times \text{Days}$ across all frequencies (OD, BD, TID, QID, STAT, Q4H, Q6H, Q8H, WEEKLY). Includes real-time cross-reactivity allergy checking with mandatory clinical override audit logging.
- **Signed Clinical Notes Immutability:** SOAP (Subjective, Objective, Assessment, Plan) and narrative notes. Once digitally signed, notes are frozen into immutable records. Amendments require audited parent-linkage.

2. Discharge Summary & Audit Log Workstation

HOSPITAL DISCHARGE & AUDIT

Routes: /MasterV9.3/dischargeSummary • /MasterV9.3/dischargeSummaryAuditLog

Primary Function: Generation, multi-disciplinary reconciliation, physician digital signing, and auditing of patient hospital discharge summaries.

- **Hospital Course Synopsis:** Summarizes admission reason, operative procedures, ICU length-of-stay, and hospital course timeline.
- **Discharge Medication Reconciliation:** Reconciles home medications, inpatient medications discontinued, and new post-discharge prescriptions with formatted dosage instructions.
- **Discharge Condition & Instructions:** Condition classification (Stable, Improved, Transferred, Against Medical Advice / LAMA), scheduled follow-up dates, wound dressing protocols, and emergency red-flag symptoms.
- **Discharge Summary Audit Log:** Complete immutable audit logging of every revision, approval, and digital signature with clinician timestamps and IP addresses.

3. EMR Form Assembly & Specialty Templates

DYNAMIC CLINICAL FORMS

Route: /MasterV9.3/emrPanelSelection • State: emrFormAssembly

Primary Function: Dynamic clinical form builder and specialty template assignment across clinical hospital departments.

- **Modular Widget Customizer:** Assembles custom consultation panels: Dental charting, Ophthalmology slit-lamp, OB/GYN antenatal records, Anesthesia pre-op assessments, and Physiotherapy range-of-motion.
- **Departmental Mapping:** Binds consultation form layouts to specific clinical departments (Cardiology, Orthopedics, Pediatrics, etc.) for streamlined specialty charting.
- **Safety Checklists & Questionnaires:** Configures surgical pre-op safety checklists (WHO standard), triage questionnaires, and informed consent documents.

4. EMR Masters & Clinical Catalogs

MASTER CONFIGURATIONS

Route: /MasterV9.3/emrMasters • Catalogs: 50+ Medical Master Tables

Primary Function: Centralized governance of standardized medical catalogs, safety thresholds, and clinical classifications.

Master Catalog	Clinical Scope & Functionality
Allergies & Reactions	Formulations, drug classes, severity levels (Severe, Moderate, Mild), and cross-reactivity rules (e.g. Penicillins $\rightarrow$ Cephalosporins).
Vaccine & Immunization	WHO/CDC immunization catalogs, vaccine brands, route, lot numbers, dosage, and booster schedules.
Lines, Drains & Tubes (LDT)	Registry of peripheral IV cannulas, Central Venous Catheters, Foley catheters, JP drains, and chest tubes.
Clinical Assessment Scales	Morse Fall Risk, Humpty Dumpty Pediatric Scale, GAD-7 Anxiety Scale, PHQ-9 Depression Scale, and Braden Ulcer Scale.
Drug Schedules & Shifts	Ward administration time windows (Morning 08:00, Afternoon 14:00, Evening 20:00, Night 02:00) and PRN dosing limits.

5. Inpatient eMAR & Ward Care Workstation

INPATIENT NURSING & MEDICATION ADMINISTRATION

Route: /MasterV9.3/patientRegistration#tabviewIPDMedication

Primary Function: Real-time 24-hour electronic medication administration, bedside nursing safety validation, and fluid balance charting.

- **24-Hour Administration Timeline Matrix:** Interactive visual matrix displaying all active inpatient prescriptions plotted across hourly windows (00:00–23:00) with color-coded statuses (DUE, GIVEN, HELD, REFUSED, MISSED).
- **5-Rights Administration Safety Gate:** Bedside verification checklist requiring confirmation of: *Right Patient, Right Drug, Right Dose, Right Route, and Right Time*.
- **Mandatory Non-Administration Reasons:** Enforces clinical reason capture for held or refused doses (e.g. *Patient NPO, Low SBP < 90, Patient Refused*).
- **Fluid Balance Chart (Intake & Output):** Calculates cumulative 24h intake (Oral, IV, Blood, NG) vs. output (Urine, Drains, Vomitus, Stool) and net balance ($\pm \text{mL}$).
- **Structured ISBAR Shift Handover:** Shift handoffs structured under **Identification, Situation, Background, Assessment, and Recommendation** with outgoing and incoming nurse signatures.

6. EMR Visit Summary & Longitudinal History

LONGITUDINAL PATIENT CHART

Route: /MasterV9.3/emrVisitSummary

Primary Function: Consolidated multi-encounter clinical history across outpatient visits, emergency admissions, and hospitalizations.

- **Multi-Encounter Timeline:** Chronological record of all past visit dates, treating doctors, admitting wards, and discharge diagnoses.
- **Longitudinal Biometrics:** Multi-year trend graphs for Blood Pressure, SpO2, HbA1c, Weight, and Lipid profiles.
- **Diagnostic Result Stream:** Direct access to historical laboratory reports, radiology imaging interpretations, and surgical procedure operative notes.

7. Clinical Governance, Audit Trail & CDS Telemetry

HIPAA & CLINICAL SAFETY GOVERNANCE

Routes: /MasterV9.3/emrUserReport • /MasterV9.3/emrVisitReport

Primary Function: Regulatory HIPAA access auditing, emergency break-glass tracking, and Clinical Decision Support (CDS) safety analytics.

- **Immutable Access & Mutation Logs:** Captures every chart view, note update, and prescription modification with clinician ID, role, client IP, timestamp, and before/after data diffs.
- **Emergency Break-Glass Audit:** Mandates and records emergency clinical justifications when clinicians override access restrictions on sensitive patient charts.
- **CDS Safety Telemetry:** Real-time dashboard tracking drug allergy warning override rates, critical vitals threshold breaches, and clinician response metrics.

3. Application Parity & Automated Verification Matrix

All 7 core EMR screens, dosing calculation rules, and clinical safety gates are verified against the application codebase:

Clinical Gate	Work Packages & Modules	Validation Scope	Test Result
Gate A	Phase 0: Compilation, CORS, Bridge	TypeScript build, callback-safe snapshotting, IST date shift fix.	PASSED (100%)
Gate B	Phase 1: Clinical Safety & Notes	Signed notes immutability, Rev optimistic locking (HTTP 409), patient context gate.	PASSED (100%)
Gate C	Phase 2: Patient Chart & Rx Pad	Identity banner, Allergy screen, Vitals & BMI, ICD-10 CPOE, Dose math with multiplier.	PASSED (100%)
Gate D	Phase 3: Ward & Inpatient eMAR	24h eMAR matrix, 5-Rights safety checklist, ISBAR handover, Fluid Balance (+/-).	PASSED (100%)

Clinical Gate	Work Packages & Modules	Validation Scope	Test Result
Gate E	Phase 4: Governance & Auditing	HIPAA audit logging, emergency break-glass tracking, CDS safety telemetry.	PASSED (100%)